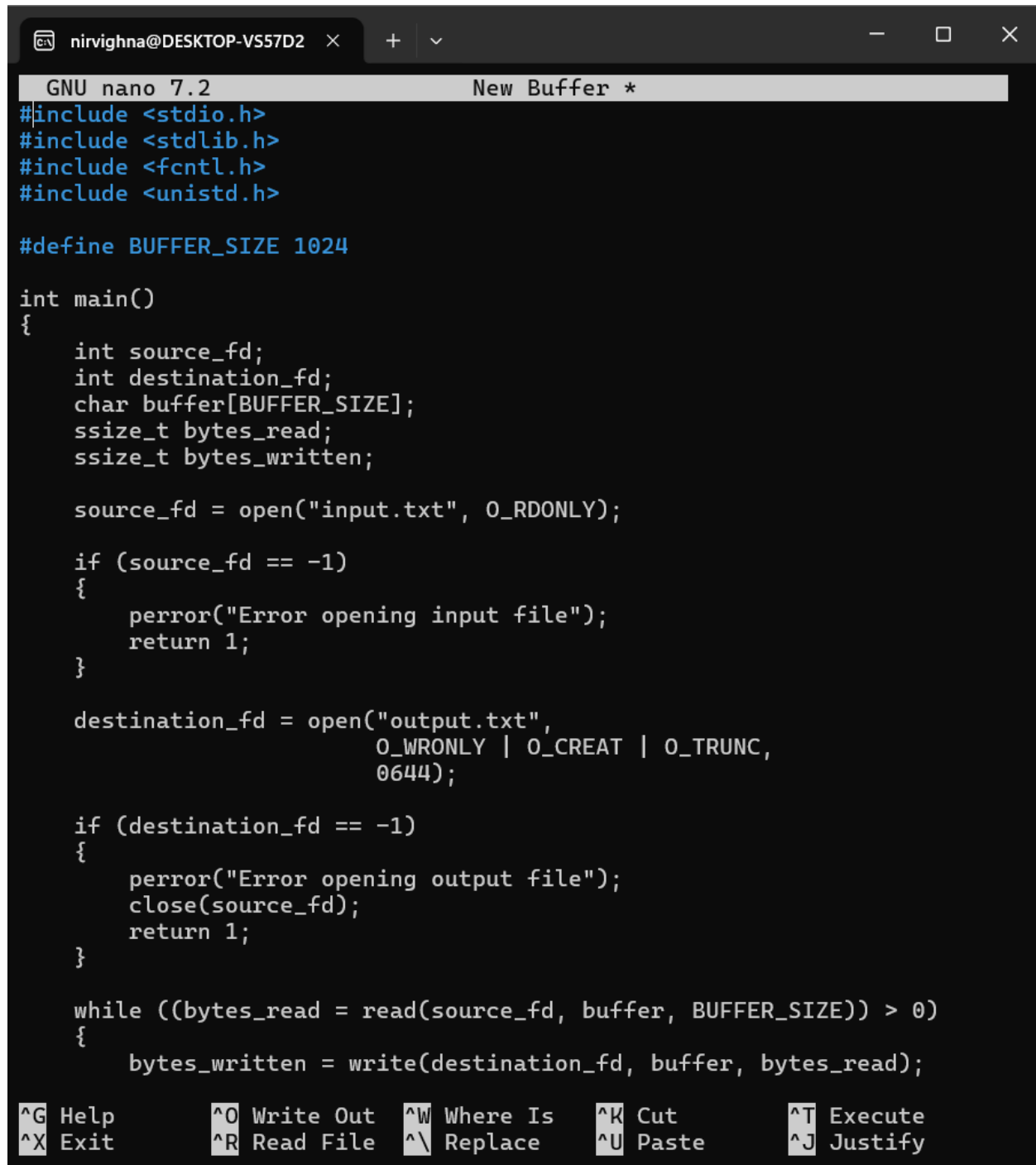


PRACTICAL-2:



The image shows a terminal window with a dark background. The title bar at the top indicates the user is 'nirvighna@DESKTOP-VS57D2'. The terminal itself shows the GNU nano 7.2 editor with a file named 'New Buffer *'. The code is a C program that reads from 'input.txt' and writes to 'output.txt' using a buffer of size 1024. The code includes standard headers, defines the buffer size, and uses file descriptors to open and read/write files. Error handling is present for both file opening and writing. The bottom of the screen shows a row of keyboard shortcuts for nano editor commands.

```
GNU nano 7.2 New Buffer *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main()
{
    int source_fd;
    int destination_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read;
    ssize_t bytes_written;

    source_fd = open("input.txt", O_RDONLY);

    if (source_fd == -1)
    {
        perror("Error opening input file");
        return 1;
    }

    destination_fd = open("output.txt",
                          O_WRONLY | O_CREAT | O_TRUNC,
                          0644);

    if (destination_fd == -1)
    {
        perror("Error opening output file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination_fd, buffer, bytes_read);
    }
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ~/OS-Week1
-bash: /home/nirvighna/OS-Week1: Is a directory
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/file_copy.c -
o bin/file_copy
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/file_copy
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 14:57 bin/file_copy
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
munmap(0x708fc2b1f000, 19695) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY) = 3
openat(AT_FDCWD, "output.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
read(3, "This is Practical Session 2 usin...", 1024) = 54
write(4, "This is Practical Session 2 usin...", 54) = 54
read(3, "", 1024) = 0
close(3) = 0
close(4) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
getrandom("\x37\xbc\xbf\xe8\x46\xe4\xf2\x06", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x61a68a8cc000
brk(0x61a68a8ed000) = 0x61a68a8ed000
write(1, "File copied successfully.\n", 26File copied successfully.
) = 26
exit_group(0) = ?
+++ exited with 0 +++
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$
```

```
+++ exited with 0 +++
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ strace -e trace=openat,read,write,clos
e ./bin/file_copy
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"
..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY) = 3
openat(AT_FDCWD, "output.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
read(3, "This is Practical Session 2 usin...", 1024) = 54
write(4, "This is Practical Session 2 usin...", 54) = 54
read(3, "", 1024) = 0
close(3) = 0
close(4) = 0
write(1, "File copied successfully.\n", 26File copied successfully.
) = 26
+++ exited with 0 +++
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
openat(AT_FDCWD, "input.txt", O_RDONLY) = 3
openat(AT_FDCWD, "output.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
read(3, "This is Practical Session 2 usin...", 1024) = 54
write(4, "This is Practical Session 2 usin...", 54) = 54
read(3, "", 1024) = 0
close(3) = 0
close(4) = 0
write(1, "File copied successfully.\n", 26) = 26
+++ exited with 0 +++
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat output.txt
This is Practical Session 2 using Linux system calls.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```